became the common property of chemists only several years later: the law of the conservation of mass. Lavoisier already claimed that in every chemical change the total mass of the substances involved remains constant—meaning that the total quantity of converted matter weighs exactly as much after the conversion as it did before it. The discovery and formulation of this law marks the actual beginning of modern chemistry, and in a very few years it became the connecting link between the chemistry of Boyle and the atomic theory of Gassendi.

In 1792, the German Richter discovered that chemical elements always combine in chemical compounds in definite quantitative proportions. It is not possible for just any arbitrary quantity of hydrogen to combine with just any arbitrary quantity of oxygen to form water: hydrogen and oxygen must always combine in the proportion of 1: 8 to form water. Otherwise, there remains an unconverted residue of hydrogen or oxygen. This *law of constant proportions* was then raised by Dalton to the status of a fundamental law of chemistry, and within a fairly short time it led to the union of chemistry with atomic theory. Dalton stated this law in a more precise form, and gave it a geometrical interpretation.

It is this very geometrical interpretation that is of paramount importance. We shall try to make it more intelligible by an example. When hydrogen combines with oxygen to form water, we must visualize this process as a mutual combination of the smallest particles—the atoms—of both elements in a higher, more complex unit, a water molecule, according to our modern terminology. We are now in a position to visualize a molecule as a geometrical structure composed of individual atoms, and a water molecule as a structure composed of two hydrogen atoms and one oxygen atom. This view makes the law of multiple proportions directly understandable. The compound which we call water is thus characterized by the 1:2 ratio of oxygen and hydrogen atoms.

This theory of Dalton, advanced in 1803, of atoms combining in molecules in a manner capable of geometrical illustration, was developed further and raised, within a few years, to the status of an established scientific postulate. As early as 1811, Avogadro announced a daring hypothesis, by which he laid the corner stone of what we know to-day as the chemical theory of atoms. He maintained that at the same temperature and pressure, equal volumes of all gases contained the same number of molecules. Although this hypothesis was still in need of experimental proof, it soon turned out to be the clue to the determination of atomic weights, and it also provided a solid, permanent foundation for Dalton’s atomic theory. If the number of atoms or molecules contained in a specific quantity of gas is known, the composition of an individual molecule can be stated exactly—for instance, whether a water molecule actually consists of one oxygen atom and two hydrogen atoms.